Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 7)

We will go through Questions 2 & 3 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. Before the launch of the space shuttle *Challenger*, information was available from 23 previous shuttle launches. For launch i, let $Y_i = 1$ if an incident of damage to the field-joint O-rings was observed and let $Y_i = 0$ otherwise. Let x_i denote the launch temperature in degrees Fahrenheit. Consider the logistic regression model

$$Y_i \sim \text{Bernoulli}(p_i), \qquad \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_i.$$

The data and model were entered in R as follows.

```
orings <- data.frame(
  temperature = c(53, 57, 58, 63, 66, 67, 67, 67, 68, 69,
                  70, 70, 70, 70, 72, 73, 75, 75, 76, 76,
                  78, 79, 81),
  damage = c(1, 1, 1, 1, 0, 0, 0, 0, 0, 0,
             1, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)
)

fit <- glm(damage ~ temperature, family = binomial,
           data = orings)
summary(fit)
vcov(fit)
```

Relevant output is shown below.

```
Call:
glm(formula = damage ~ temperature, family = binomial,
    data = orings)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.0611  -0.7613  -0.3783   0.4524   2.2175

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   15.0429     7.3786   2.039   0.0415
 temperature  -0.2322     0.1082  -2.145   0.0320

(Dispersion parameter for binomial family taken to be 1)

> vcov(fit)
              (Intercept) temperature
(Intercept)     54.444183   -0.796385
 temperature    -0.796385    0.011715
```

Let T_{50} be the launch temperature at which the model gives a probability of 0.5 that an O-ring damage incident occurs.

(a) Express T_{50} in terms of β_0 and β_1. Hence, determine its maximum likelihood estimate and state the property of maximum likelihood estimators used.
Solution. At T_{50}, the probability of damage is 0.5, and hence the log-odds are zero. Therefore,

$$0 = \beta_0 + \beta_1 T_{50}, \qquad \text{so} \qquad T_{50} = -\frac{\beta_0}{\beta_1}.$$

By the invariance property of maximum likelihood estimators,

$$\widehat{T}_{50} = -\frac{\widehat{\beta}_0}{\widehat{\beta}_1} = -\frac{15.0429}{-0.2322} = 64.8^\circ\text{F}.$$

(b) Suppose that $\widehat{\boldsymbol{\theta}}$ is an MLE of $\boldsymbol{\theta} \in \mathbb{R}^d$ and that $g : \mathbb{R}^d \longrightarrow \mathbb{R}$ is differentiable. Recall that

$$g(\widehat{\boldsymbol{\theta}}) \mathrel{\dot\sim} N\Big(g(\boldsymbol{\theta}), \nabla g(\boldsymbol{\theta})^\mathsf{T} \mathcal{I}(\widehat{\boldsymbol{\theta}})^{-1} \nabla g(\boldsymbol{\theta})\Big),$$

where $\mathcal{I}(\boldsymbol{\theta})$ is the expected information matrix. Use this result and the output from `vcov(fit)` to calculate the estimated standard error of $\widehat{T}_{50}$.
Solution. Define

$$g(\beta_0, \beta_1) = -\frac{\beta_0}{\beta_1}.$$

Then

$$\nabla g(\beta_0, \beta_1) = \begin{pmatrix} -1/\beta_1 \\ \beta_0/\beta_1^2 \end{pmatrix}.$$

Evaluated at the MLE,

$$\nabla g(\widehat{\beta}_0, \widehat{\beta}_1) = \begin{pmatrix} -1/(-0.2322) \\ 15.0429/(-0.2322)^2 \end{pmatrix} = \begin{pmatrix} 4.307 \\ 279.091 \end{pmatrix}.$$

The matrix reported in `vcov(fit)` estimates $\mathcal{I}(\widehat{\boldsymbol{\beta}})^{-1}$. Consequently,

$$\begin{aligned} \widehat{\text{Var}}(\widehat{T}_{50}) &= \begin{pmatrix} 4.307 & 279.091 \end{pmatrix} \begin{pmatrix} 54.444183 & -0.796385 \\ -0.796385 & 0.011715 \end{pmatrix} \begin{pmatrix} 4.307 \\ 279.091 \end{pmatrix} \\ &\approx 7.891. \end{aligned}$$

Thus,

$$\widehat{\text{se}}(\widehat{T}_{50}) = \sqrt{7.891} = 2.81^\circ\text{F}.$$

(c) Suppose temperature is instead measured in degrees Celsius, with

$$C = \frac{5}{9}(T - 32).$$

Express the logistic regression model in terms of C. Determine how the fitted intercept, fitted slope, $\widehat{T}_{50}$ and its estimated standard error change. Explain why the fitted probabilities do not change.
Solution. Since $T = 32 + \frac{9}{5}C$,

$$\begin{aligned} \text{logit}(p) &= \beta_0 + \beta_1 \left(32 + \frac{9}{5}C\right) \\ &= (\beta_0 + 32\beta_1) + \frac{9}{5}\beta_1 C. \end{aligned}$$

Thus, for the model written as $\text{logit}(p) = \beta_0^* + \beta_1^* C$,

$$\widehat{\beta}_0^* = \widehat{\beta}_0 + 32\widehat{\beta}_1 = 7.6137, \qquad \widehat{\beta}_1^* = \frac{9}{5}\widehat{\beta}_1 = -0.4179.$$

The estimated 50% temperature on the Celsius scale is

$$\widehat{C}_{50} = \frac{5}{9}(\widehat{T}_{50} - 32) = 18.22^\circ\text{C}.$$

Because the conversion from Fahrenheit to Celsius multiplies temperature differences by 5/9,

$$\widehat{\text{se}}(\widehat{C}_{50}) = \frac{5}{9}\widehat{\text{se}}(\widehat{T}_{50}) = 1.56^\circ\text{C}.$$

The fitted probabilities do not change because the re-expressed coefficients give the same fitted linear predictor at every physical temperature.

2. The `kyphosis` data frame contains information on 81 children who underwent corrective spinal surgery. The response records whether kyphosis was present or absent following surgery. The explanatory variables are the child's age in months (`Age`), the number of vertebrae involved in the operation (`Number`), and the number of the first, or topmost, vertebra operated on (`Start`). Define a successful outcome to mean that kyphosis was absent following surgery.
The following logistic regression model was fitted in R.

```
data(kyphosis, package = "rpart")
kyphosis$success <- as.integer(kyphosis$Kyphosis == "absent")
fit <- glm(success ~ Age + Number + Start,
           family = binomial, data = kyphosis)
summary(fit)
```

Relevant output is shown below.

```
Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   2.03693    1.44958   1.405  0.16003
Age          -0.01093    0.00645  -1.696  0.08996
Number       -0.41060    0.22486  -1.826  0.06791
Start         0.20651    0.06771   3.050  0.00229

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 83.234  on 80  degrees of freedom
Residual deviance: 61.380  on 77  degrees of freedom
AIC: 69.380
```

Let p_i denote the probability of a successful outcome for child i.

(a) Write down the fitted logistic regression model.
Solution. The fitted model is

$$\log\left(\frac{\widehat{p}_i}{1-\widehat{p}_i}\right) = 2.03693 - 0.01093\,\text{Age}_i - 0.41060\,\text{Number}_i + 0.20651\,\text{Start}_i.$$

(b) Interpret the fitted coefficient for `Start` in terms of the odds of a successful outcome.
Solution. For children of the same age and with the same number of vertebrae involved, increasing `Start` by one vertebra multiplies the fitted odds of a successful outcome by

$$\exp(0.20651) = 1.229.$$

Thus, the fitted odds of postoperative kyphosis being absent are about 1.23 times as large when the first vertebra operated on increases by one, holding `Age` and `Number` fixed.

(c) Use the fitted coefficient for `Age` to determine the multiplicative change in the odds of a successful outcome associated with an increase in age of one year, holding `Number` and `Start` fixed. Interpret the result.

Solution. Age is measured in months, so an increase of one year corresponds to an increase of 12 in `Age`. The fitted odds are multiplied by

$$\exp(12 \times -0.01093) = 0.877.$$

Thus, for fixed values of `Number` and `Start`, the fitted odds of a successful outcome are multiplied by about 0.88 for each additional year of age. Equivalently, the fitted odds are about 12.3% lower.

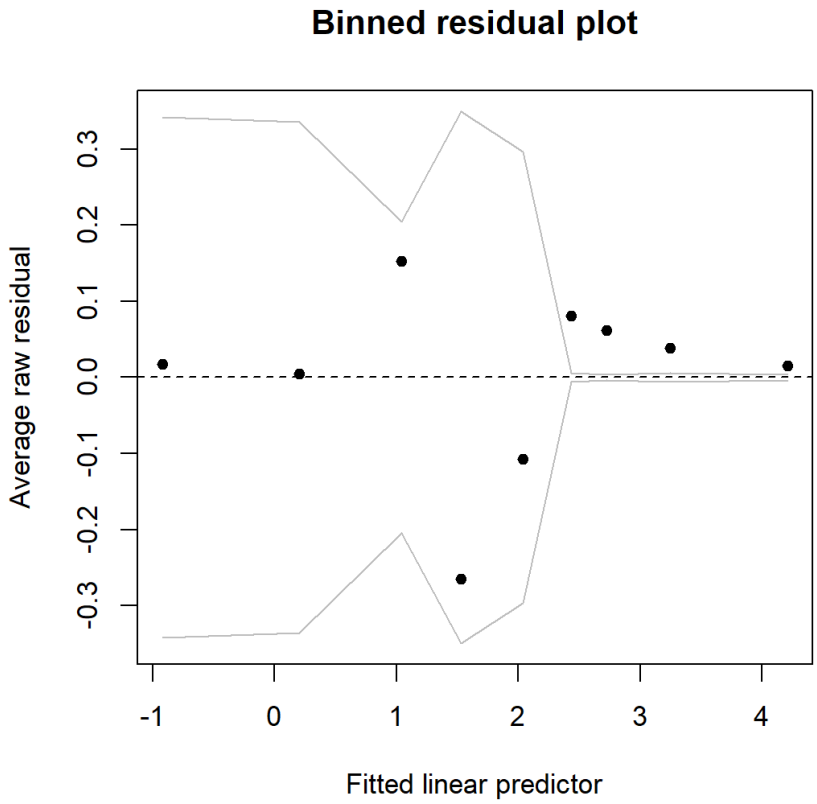
(d) Explain why a plot of the individual raw residuals $y_i - \widehat{p}_i$ against the fitted linear predictors is difficult to interpret for this model. How does binning the residuals make the plot more informative?

Solution. Because y_i is binary, the raw residual is

$$y_i - \widehat{p}_i = \begin{cases} 1 - \widehat{p}_i, & y_i = 1, \\ -\widehat{p}_i, & y_i = 0. \end{cases}$$

The individual residuals therefore lie on two curved branches determined by the binary response, which makes systematic departures from the fitted model difficult to see. Averaging residuals within intervals of the fitted linear predictor reduces this binary pattern. Under an adequate model, the binned average residuals should be scattered around zero without a systematic pattern.

(e) A binned residual plot for the fitted model is shown below. The grey lines are calculated as approximately two standard errors above and below zero, where the standard error in each bin is estimated using the sample standard deviation of the raw residuals in that bin. For some bins at the right of the plot, the observed responses are nearly all the same. The raw residuals in such a bin consequently have very little empirical variation, causing the grey reference limits to collapse towards zero. Points outside these very narrow limits should therefore not automatically be interpreted as evidence of lack of fit. What does the plot suggest about the adequacy of the fitted model? Justify your conclusion by considering the overall pattern of the binned residuals, as well as the behaviour of the grey reference limits.



The plot was produced using the following code. This code will be useful in Question 3.

```
library(arm)
binnedplot(x = predict(fit, type = "link"),
           y = residuals(fit, type = "response"),
           nclass = 9,
           xlab = "Fitted linear predictor",
           ylab = "Average raw residual")
```

Solution. There is no clear systematic pattern in the binned residuals. The relatively large positive and negative residuals in the middle of the plot remain within the reference limits. At the right of the plot, the residuals are all positive, but they are small and only appear unusual because the empirical reference limits have collapsed. This may indicate slight underprediction for observations with large fitted probabilities, but does not provide clear evidence of lack of fit.

(f) Estimate the probability of a successful outcome for a child aged 96 months when six vertebrae are involved in the operation and the first vertebra operated on is vertebra 8.
Solution. The fitted linear predictor is

$$\begin{aligned}\widehat{\eta} &= 2.03693 - 0.01093(96) - 0.41060(6) + 0.20651(8) \\ &\approx 0.1761.\end{aligned}$$

Therefore, the estimated probability of a successful outcome is

$$\widehat{p} = \frac{\exp(0.1761)}{1+\exp(0.1761)} \approx 0.544.$$

(g) Suppose the response were instead coded as 1 when postoperative kyphosis is present. Explain how the fitted regression coefficients and fitted probabilities would change.

Solution. If $p_i^* = 1 - p_i$ denotes the probability that kyphosis is present, then

$$\log\left(\frac{p_i^*}{1-p_i^*}\right) = -\log\left(\frac{p_i}{1-p_i}\right).$$

Consequently, every fitted regression coefficient changes sign. The fitted probability of kyphosis being present is $1 - \widehat{p}_i$, so the fitted probabilities for the two possible outcomes add to one.

3. The `Wells` data contain information from 3020 households in Bangladesh whose wells had unsafe arsenic levels. The response `switch` records whether a household switched to another well. The explanatory variables considered here are distance to the nearest known safe well, arsenic concentration in the household's original well, and years of education of the household head. Load and prepare the data as follows.

```
data(Wells, package = "carData")
Wells$switch01 <- as.integer(Wells$switch == "yes")
Wells$dist100 <- Wells$distance/100
Wells$educ4 <- Wells$education/4
```

Thus, `dist100` measures distance in hundreds of metres and `educ4` measures education in units of four years.

(a) Fit a logistic regression model for switching wells using `dist100`, `arsenic`, and `educ4` as explanatory variables. Store the fitted model as `fit.wells` and examine the fitted coefficients.
Solution.

```
fit.wells <- glm(switch01 ~ dist100 + arsenic + educ4,
                 family = binomial, data = Wells)
summary(fit.wells)
```

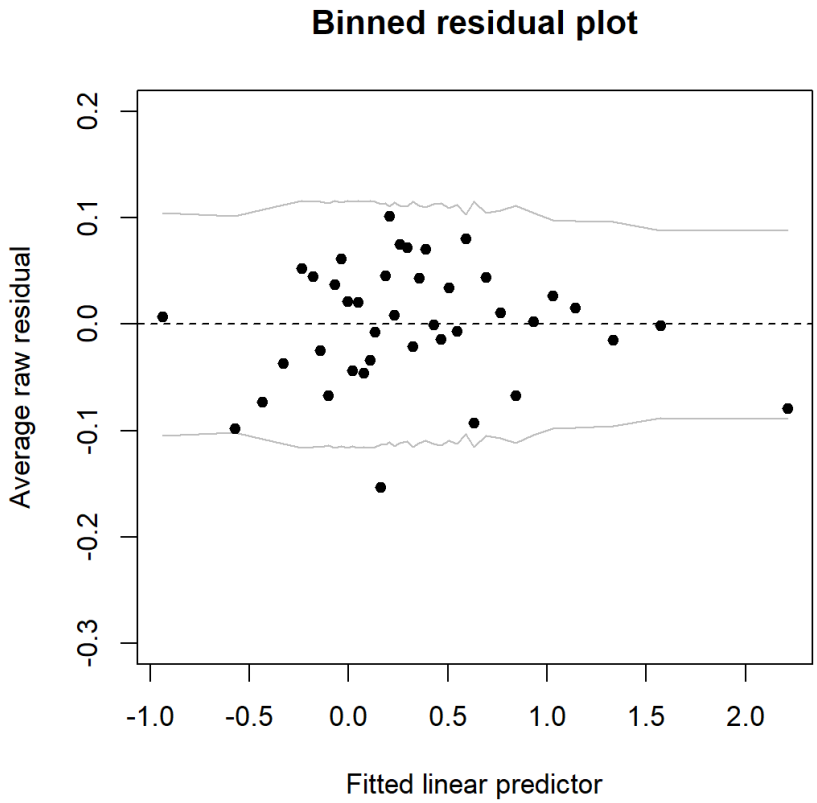
The estimated coefficients are approximately -0.214, -0.896, 0.468, and 0.171 for the intercept, distance, arsenic concentration, and education, respectively.

(b) Interpret the estimated coefficient of `dist100` in terms of the odds of switching wells.
Solution. Holding arsenic concentration and education fixed, an additional 100 metres to the nearest safe well multiplies the fitted odds of switching by $\exp(-0.896) \approx 0.408$. Thus the fitted odds are about 59.2% lower for each additional 100 metres.

(c) Interpret the estimated coefficient of `arsenic` in terms of the odds of switching wells.
Solution. Holding distance and education fixed, an increase of one unit in arsenic concentration multiplies the fitted odds of switching by $\exp(0.468) \approx 1.60$. Thus the fitted odds are about 60% higher for a one-unit increase in arsenic concentration.

(d) Using the code in Question 2(e) as a guide, produce a binned residual plot of the raw residuals against the fitted linear predictors. Use 40 bins. Does this plot provide clear evidence of lack of fit? Explain.
Solution. One possible command is

```
binnedplot(x = predict(fit.wells, type = "link"),
           y = residuals(fit.wells, type = "response"),
           nclass = 40,
           xlab = "Fitted linear predictor",
           ylab = "Average raw residual")
```

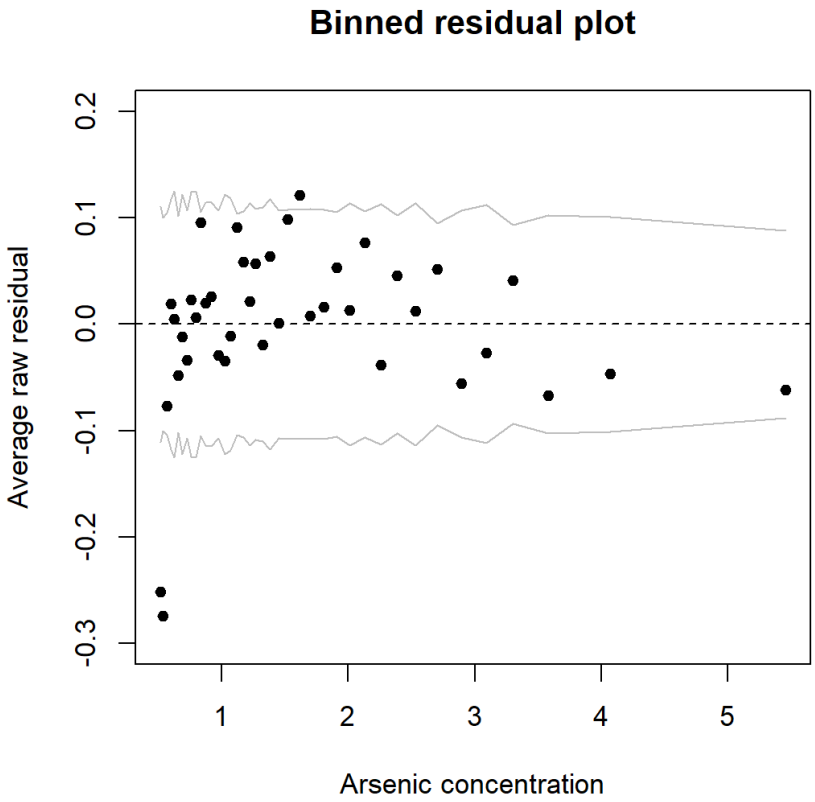


The plot does not provide clear evidence of lack of fit. The binned residuals are scattered on both sides of zero, with no clear sustained trend or curvature. A small number of points near or just beyond the reference limits would not, by themselves, constitute clear evidence of a systematic discrepancy.

(e) Produce a binned residual plot of the raw residuals against `arsenic`, again using 40 bins. Interpret the plot.

Solution. One possible command is

```
binnedplot(x = Wells$arsenic,
           y = residuals(fit.wells, type = "response"),
           nclass = 40,
           xlab = "Arsenic concentration",
           ylab = "Average raw residual")
```



The two binned residuals corresponding to the lowest arsenic concentrations lie far below the lower grey reference line. Thus, for households with the lowest arsenic concentrations, the fitted model substantially overestimates the probability of switching wells.
There is also a broader systematic pattern in the residuals. After the two large negative residuals at the lowest arsenic concentrations, the residuals tend to be positive over the next part of the arsenic range and become more negative again at higher concentrations. The relationship between arsenic concentration and the probability of switching is therefore not adequately represented across the observed range by the fitted model.

(f) What aspect of the fitted model should be reconsidered in light of the plot in part (e)?
Solution. The form of the arsenic effect should be reconsidered. The model assumes that arsenic concentration has a linear effect on the log-odds of switching, but the residual pattern suggests that this assumption is inadequate. A different functional form for arsenic should be considered; this plot alone does not determine which form should be used.